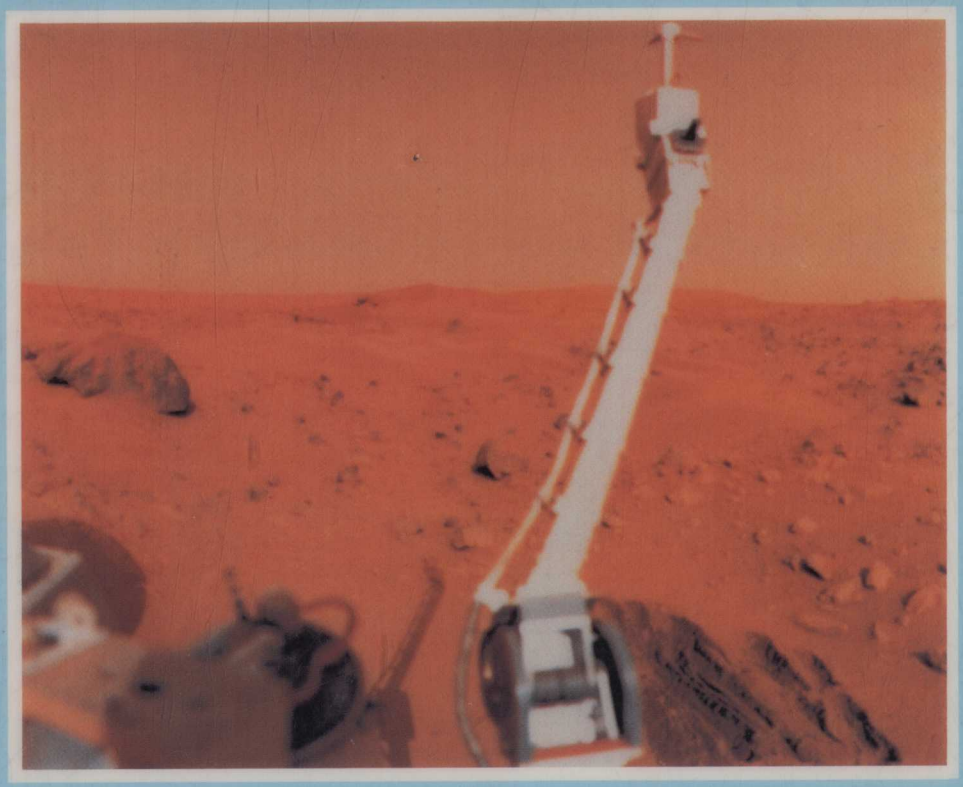
8
ST

THE NASA MARS CONFERENCE

Edited by
Duke B. Reiber



Volume 71

SCIENCE AND TECHNOLOGY SERIES

AN AMERICAN ASTRONAUTICAL SOCIETY PUBLICATION

THE METEOROLOGY OF MARS
[NMC-1G]

Dr. Conway Leovy
University of Washington

Although tenuous, the martian atmosphere can support dynamic meteorological processes. It has played an enormous role in shaping the planet's surface over time, and is the one other atmosphere in the solar system most likely to be meteorologically similar to Earth's. The southern summer pattern of winds due to dust storms reveals that most have a localized origin at low latitudes in the northern hemisphere. These move across the equator and then apparently bend eastward in a pattern analogous to a monsoon-like circulation in Earth's atmosphere. Global dust storms tend to occur when the planet is receiving maximum solar radiation near perihelion. Atmospheric pressure was found to vary significantly with the seasons. The annual minimum is recorded during the period of maximum CO_2 condensation at the South Pole (southern winter), and a weaker minimum is recorded during northern winter. Annual cycling repeats with little or no variation, but there are substantial variations in individual cycles over a period of years. Traveling cyclones and fronts were observed from orbit and were also detected by lander instruments, and lander imaging provided evidence of increased atmospheric opacity during such events. While the martian atmosphere is in many respects a laboratory for studying the meteorological processes associated with planetary atmospheric dynamics in general, there is particular interest in the cycling of water vapor between the martian atmosphere and surface. Future mission science should contribute important new knowledge to our understanding of this process.

INTRODUCTION: THE ATMOSPHERE AND ITS METEOROLOGY

The *NASA Mars Conference* presentation sequence is organized such that mine is at exactly the right place on the agenda: between Michael McElroy's [NMC-1F], which discusses that part of the atmosphere that has escaped and the resulting implications for the current atmosphere, and Fraser Fanale's [NMC-1H], which is concerned with that part of the atmosphere that has been retained in reservoirs on and in the surface. I am going to discuss only that part of the atmosphere that is actually in the atmosphere at the present time. This amounts to only two percent (2%) of Earth's atmosphere in terms of mass per unit area.

Relevancy of Martian Atmosphere

One might wonder why a meteorologist would be interested in an atmosphere that is so thin. There are a number of reasons. One important reason is that it is enough atmosphere, as on Earth, to play a role in shaping the surface. As explained and illustrated during other conference presentations, the martian atmosphere has in fact played an enormous role in shaping the planet's surface over time. However, just how that role has operated still remains, in many respects, to be deciphered.

The martian atmosphere is also of interest to meteorologists because it is the atmosphere in the solar system most likely to be similar in its dynamical processes to the atmosphere of Earth. Therefore, if we want to extend our Earth-bound views of weather systems or of atmospheric dynamics in general, Mars is the place to study next because it stretches our imaginations. But it does not stretch them so far that we find ourselves on extremely thin ice. Its processes are not nearly as difficult to understand as those driving the atmospheres of Venus or very large planets like Jupiter and Saturn.

Integration of Historic and Modern Work

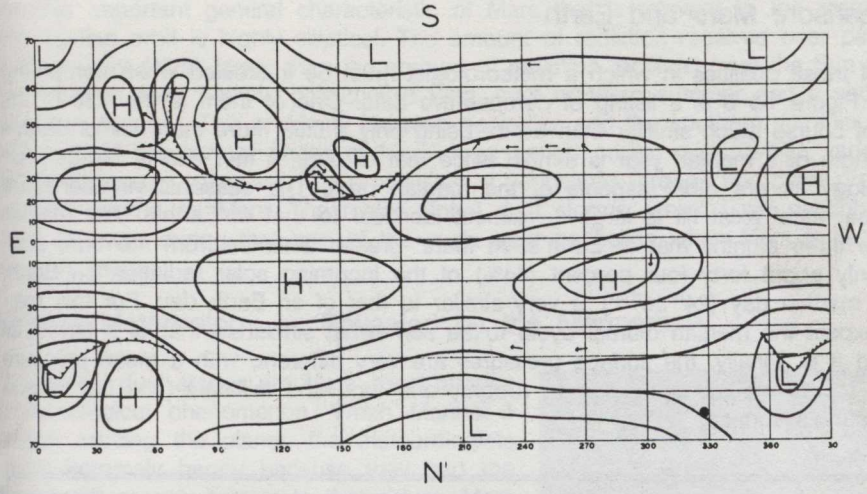
One of the first meteorologists to take a serious look at the atmosphere of Mars was Seymour Hess[1]. In 1950, Dr. Hess spent some time at the Lowell Observatory working with temperature data acquired by Coblentz and Lampland in the 1920's. These were very difficult observations to make at that time and represented temperature distributions on the surface of Mars. Dr. Hess used these data to infer distributions of atmospheric pressure, and he produced the crude surface weather map presented in Figure 1G-1. He combined several bits of ephemeral data, primarily the temperature data and, secondarily, wind data. The latter were obtained by assuming that observed drifts of yellow clouds[2] and other martian clouds corresponded with wind drift.

Figure 1G-2 presents a somewhat later example of compilations of apparent wind drift measurements at the surface of Mars, as documented by Gifford in the mid 1950's. As in the case of the data used by Hess, Gifford's data also represented Earth-based observatory measurements of the drift of yellow clouds. While it was not noted at the time, Gifford's work already revealed a very systematic pattern of winds. The southern summer pattern of wind drifts due to dust storms reveals that most of the storms originate in a localized band of longitude at low latitudes in the northern hemisphere during the southern summer. Most of these dust clouds move across the equator and then apparently bend eastward, following the wind drift. This could be very analogous to a monsoon-like circulation in the Earth's atmosphere. On Earth, during northern hemisphere winter, a monsoon circulation flows southward across the equator from the Asian sector, then bends eastward toward Australia. If we had recognized it at the time, we might have understood that a similar process was occurring in the martian atmosphere. Only recently have we begun to see the older circulation data as evidence of typical monsoon behavior for large martian dust storms.

---

[1] Until his death, Dr. Seymour L. Hess was a highly respected and noted authority on the meteorology of Mars. His work, which matured while he was in the Department of Meteorology at Florida State University, earned him the position of principal investigator and team leader for the **Viking** Meteorology Science Team. {Ed.}

[2] Prior to the detailed knowledge of Mars afforded by **Mariner 9** and **Viking** data, martian clouds were often categorized according to color. Yellow clouds were believed to be dust clouds close to the surface, and therefore most indicative of near-surface meteorological conditions like wind. **Mariner 9** and **Viking** observations dramatically demonstrated that this assumption was correct.



A schematic streamline map for Mars in northern hemisphere winter. This is the telescopic view with south at the top. To obain the usual meteorological view, merely invert the page. The arrows represent observed cloud drift directions.

Figure 1G-1: Early Map Based on Temperature and Wind Data by Seymour Hess (J. Met., 1950)

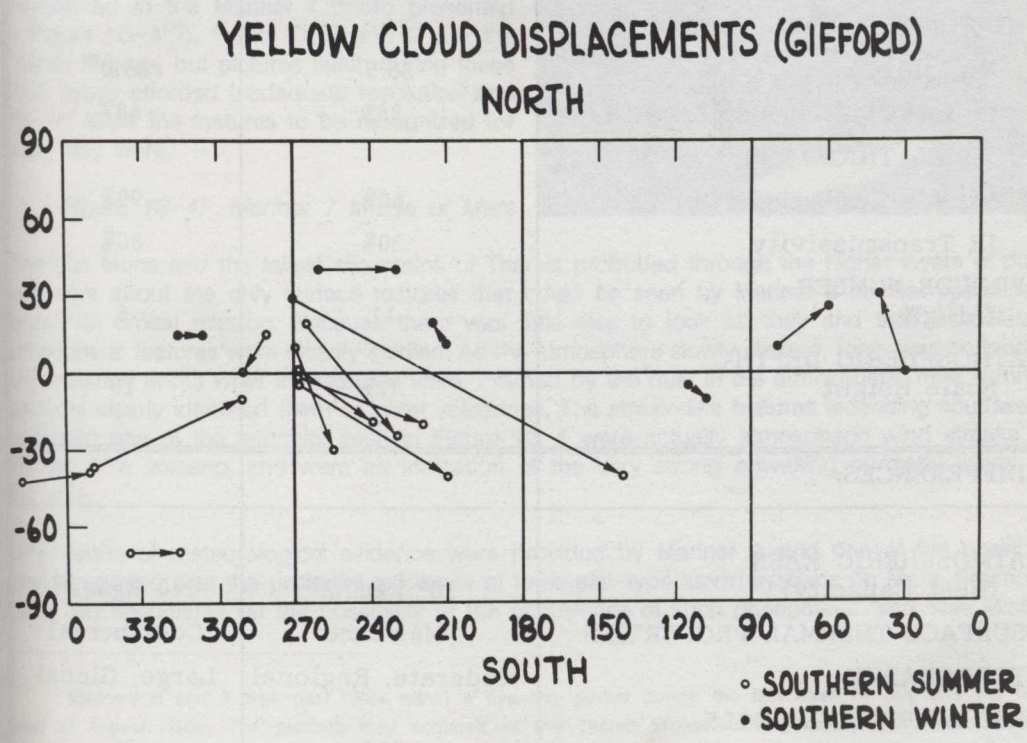


Figure 1G-2: Early Map of Wind Drift Measurements

General Comparison: Mars and Earth

As a reminder of those qualities in which a meteorologist must be interested when comparing Earth and Mars, Figure 1G-3 is a listing of comparative data. One of them is the size of the planet. Mars is of course much smaller than Earth, being only a little more than half of Earth's diameter. The length of a martian year is almost twice that of ours, a fact that is not as crucial for meteorology as are other aspects of the martian orbit. The seasonal variations are important because Mars' axial tilt is large -- almost identical to that of Earth's. The martian surface gravity is three-eighths that of Earth's. At Mars' greater distance from the Sun, sunlight averages only about forty-four percent (44%) of the incoming solar radiation on Earth. The length of a martian day ($24^{hr}38^{min}$) is very similar to that of an Earth day. For this reason, we might expect the martian diurnal cycle to be somewhat similar to that of a terrestrial desert, as indeed it is. Finally, the surface pressures are very different, with a mean pressure on Mars of only seven millibars compared to roughly one-thousand millibars (one bar or one atmosphere) at Earth's surface.

SIMILARITIES:		
	EARTH	MARS
ROTATION Length of Day (sec)	86400	88640
SEASONS		
Obliquity	23.5°	25.0°
Solar Orbit Period (Earth days)	365	687
OPTICAL THICKNESS		
Visible Transmissivity	60%	90%
IR Transmissivity	30%	80%
FROUDE NUMBER f^2d^2/gH	11	6
ATMOSPHERIC DEPTH Scale Height	7.5	10
DIFFERENCES:		
ATMOSPHERIC MASS (Heat Capacity)	10^4 kgm/m^2	140 kgm/m^2
SURFACE THERMAL PROPERTIES	Maritime	Continental
TOPOGRAPHY	Moderate, Regional	Large, Global
RADIATIVE TIME SCALE $f\tau_r$	300	30

Figure 1G-3: Comparison of Earth and Mars

Another important general characteristic of Mars that is relevant to the meteorologist is that the martian orbit is highly elliptical. The amount of radiation received near perihelion is about forty percent (40%) larger than the amount of radiation received from the Sun at aphelion. This variability in solar heating produces a long, cool northern summer and a short, relatively hot southern summer. This proximity between perihelion and the southern summer solstice is characteristic of the current phase of Mars' long-period orbital cycle, which varies with a period of approximately 10^5 years. The precession of perihelions through that cycle produces important climatic cycling on Mars. The major global dust storms, which I will describe in detail later, tend to occur during that part of the cycle near perihelion when Mars is receiving maximum solar radiation.

Martian Meteorology as Revealed by the Mariners

The global dust storms are Mars' most famous meteorological phenomenon. When **Mariner 9** began orbiting the planet, the meteorologists were extremely happy because they had the spacecraft camera system to themselves. Mars was engulfed in a massive global dust storm and nobody could see the surface. The only visible features were some curious looking (apparent) mountain tops that correlated with features observed earlier by Mariners 6 and 7 (highlighted in the **Mariner 7** photo presented in Figure 1G-4[3]). Points O, P and Q are the Tharsis Montes, but pictures taken during these 1969 flybys afforded inadequate resolution and did not allow the features to be recognized for what they were.

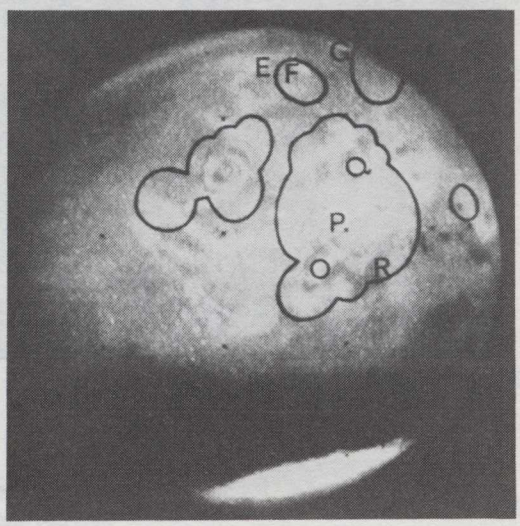


Figure 1G-4: Mariner 7 Image of Mars

Olympus Mons and the tallest mountains of Tharsis protruded through the higher layers of dust and were about the only surface features that could be seen by **Mariner 9** as that spacecraft began its orbital mission. Because there was little else to look at, they and their associated atmospheric features were closely studied. As the atmosphere slowly cleared, there was no longer any question about what the features were. Framed by the dust in the atmosphere, their summit calderas clearly identified them as giant volcanoes. The streak-like features extending southward and from one of the summits seen in Figure 1G-5 were actually atmospheric wind streaks in the lee of a volcano, and were an indication of the very strong prevailing northerly winds at the time.

Other kinds of meteorological evidence were provided by **Mariner 9**, and one of the possibilities suggested was the probable existence of terrestrial-type storm systems on Mars. Seymour Hess had speculated on the possibility of the occurrence of such phenomena, and Yale Mintz

---

[3] Mariners 6 and 7 flew past Mars within a five-day period during the last week of July and the first week of August, 1969. The pictures they acquired of the Tharsis region were generally from more distant approach ranges (low resolution). The most striking of these were frames 7F73 and 7F74, taken by **Mariner 7** on August 4, the first to reveal Olympus Mons (outlined "bull's eye" feature) and the Tharsis Montes, although with insufficient resolution to determine what they were. It wasn't until **Mariner 9**'s mission that brightnesses associated with these features were recognized to be mid-day clouds forming along elevated crests. {Ed.}

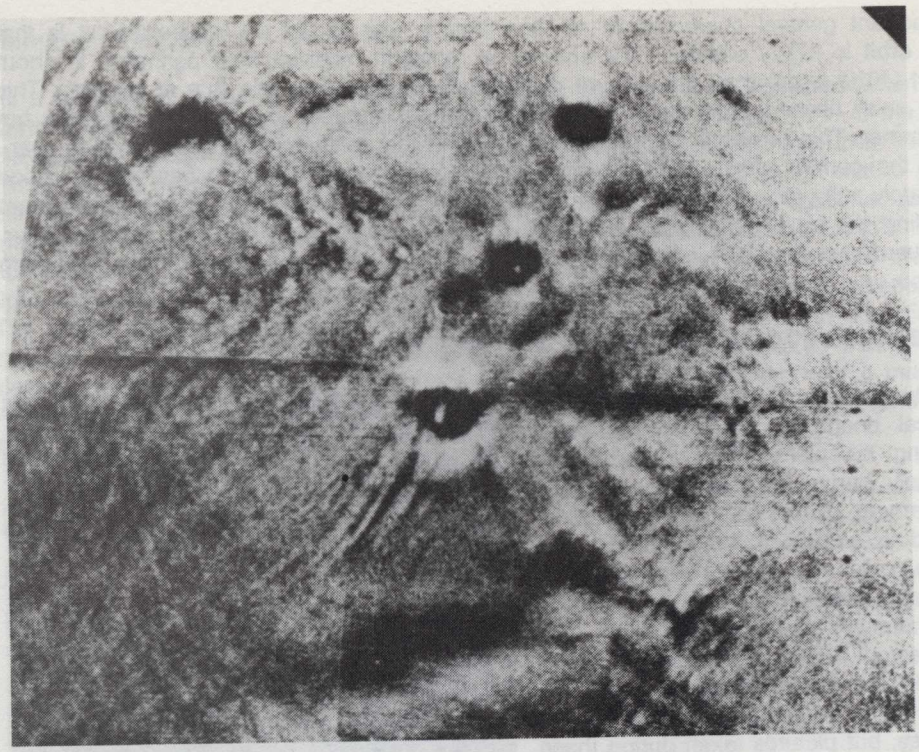


Figure 1G-5: Early Mariner 9 Image During Global Dust Storm

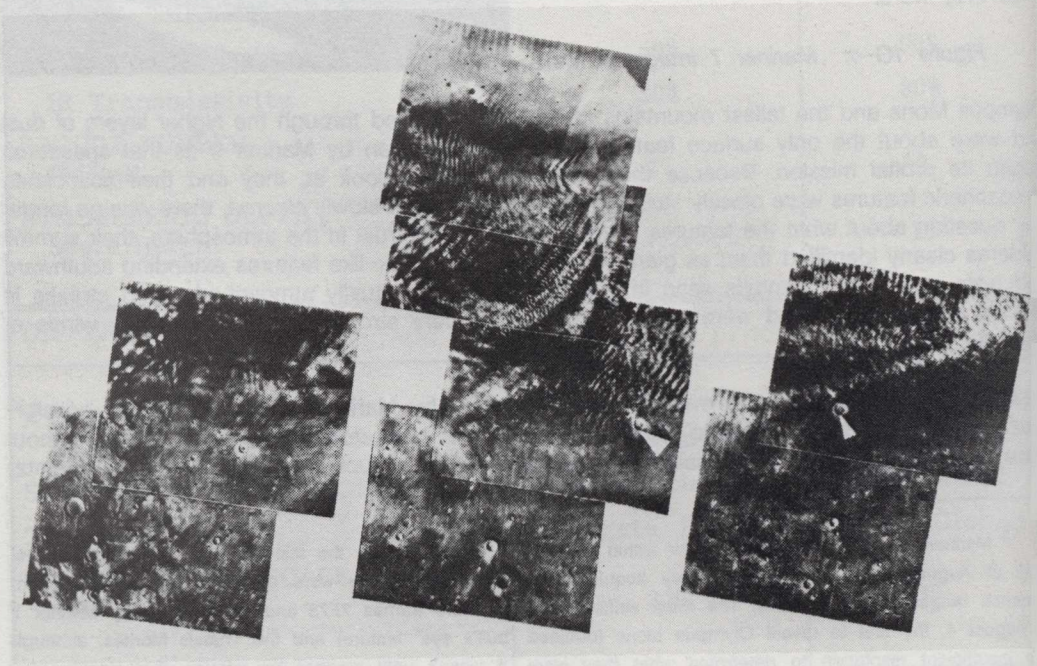


Figure 1G-6: Progress of Cold Front Imaged by Mariner 9 (each mosaic 1500 km acro

[1976a] later did a detailed analysis of the possibility. The implication of Mintz's work was that weather systems (such as fronts embedded in moving cyclones) were likely in the middle latitudes during winter.

Figure 1G-6 (opposite page) presents a series of **Mariner 9** pictures (taken Feb. 11-14, 1972, with longitude and latitude lines included for 45°N, 145°W). They illustrate the progress of a feature that looks very much like a terrestrial cold front. The front is the bright band extending from the lower left to the upper right just above the middle of the central mosaic. On the second day of this sequence, there is evidence of an intense dust storm associated with strong northerly winds in the lower-left portion of the frame. On the third day of the sequence, a large crater rim is seen to be producing wave clouds (presumably composed of water ice) that resemble a sonic boom's shock wave. The clouds were being produced by extremely strong low-level winds passing over the crater, and the day-to-day variations are indicative of day-to-day weather changes and frontal systems. These weather extremes were observed by **Mariner 9** following the global dust storm described earlier.

Viking Meteorology Instrumentation

Figure 1G-7 is an illustration of the primary **Viking** lander meteorology instrument. The instrument package was located at the end of an arm-like boom that was equipped with an elbow hinge so that the instrument could be deployed from its stowed position after landing; it is in fact clearly visible in some of the **Viking** lander pictures taken at the surface of Mars. The device consisted of thermocouple units to measure the atmospheric temperature, and wind speed sensors. Each lander also had an atmospheric pressure sensor which, because it had to be shielded from wind, was mounted inside the lander rather than in the boom assembly.

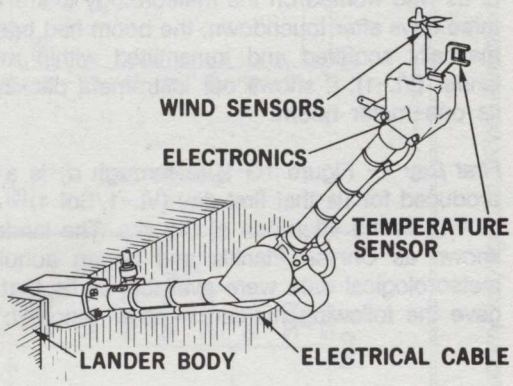


Figure 1G-7: Viking Meteorology Sensors

Measuring wind speed required the use of two orthogonally oriented, heated sensors, together with an unheated reference-temperature sensor. The power required to maintain a constant overheat of the two orthogonal sensors relative to the reference is measured, and from this the cooling due to wind--and thence the wind speed itself--can be determined. One cannot uniquely determine the wind direction with two orthogonal sensors, so a thermocouple array (hot-film anemometer) was used to determine the wind directional characteristics. It worked very much like a wet finger: heat it, expose it to wind, and then measure which side is cooler. Wind speed could also be crudely determined from the magnitude of the cooling. This meteorology package was mounted on each of the landers, and, for the most part, they performed quite well on the martian surface[4].

---

[4] The **Viking** landers were so well instrumented that some of the instruments did double duty, helping out other areas of science when possible. Another thermocouple sensor was mounted on one of each lander's footpads. Its primary function was to help complete an atmospheric temperature profile during the landing and then to support physical properties science by measuring temperatures at the surface boundary or in the soil if the footpad was buried (as it was with VL-1); this sensor was also helpful to meteorology. It was also found that the seismometer on VL-2 (VL-1's seismometer failed to uncage and could not be used) could provide some rough but reliable data on winds. {Ed.}

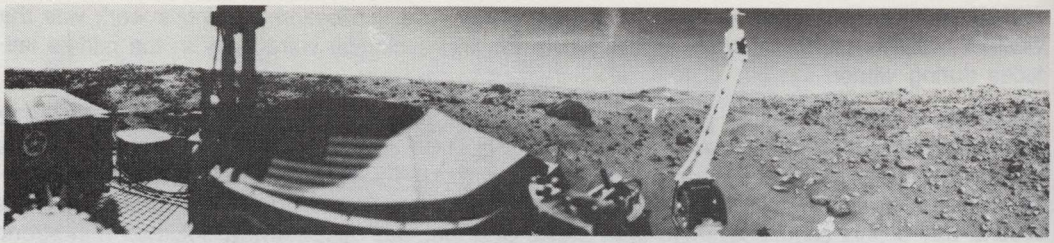


Figure 1G-8: VL-1 Panorama at Chryse Planitia Revealing Deployed Meteorology Boom

Initial Viking Meteorology Results

The picture presented in Figure 1G-8 (above) provokes a strange emotional response for those of us who worked on the meteorology team. This remarkable picture was produced by camera·1 three days after touchdown; the boom had been similarly imaged in the camera·2 panorama automatically acquired and transmitted within minutes of the historic landing by the first Viking lander (VL-1). it shows our instrument package deployed on the surface of Mars at the end of its one-meter boom.

First Day -- Figure 1G-9 (*a* through *d*) is a representation of the meteorology data that VL-1 produced for us that first day (VL-1/Sol 1)[5]. VL-1 landed July 20, 1976, at a time comparable to 4:13 in the afternoon at the site. The lander touched down at 22.27°N, 47.97°W in a basin known as Chryse Planitia and began acquiring data almost immediately. The first reduced meteorological data were available to us that same day at 5:30 pm, PDT. Seymour Hess then gave the following martian weather report[6]:

> *Light winds from the east in the late afternoon,*
> *changing to light winds from the southwest after midnight.*
> *Maximum winds were 15 miles per hour.*
> *Temperature ranged from minus 122 degrees Fahrenheit*
> *just after dawn to minus 22 degrees Fahrenheit.*
> *Pressure steady at 7.7 millibars.*

The data produced reflect the temperature (Fig. 1G-9a), wind speed (Fig. 1G-9b), wind direction (Fig. 1G-9c), and pressure (Fig. 1G-9d). These are of course the most conventional meteorological quantities, but *un*conventional instrumentation was required to be able to measure them under conditions occurring on Mars. No one knew with certainty exactly what those conditions would be, of course, and the instruments therefore had to work in an environment that could only be theorized in advance.

---

[5] A "sol" is a martian day; VL-1/Sol 1 is the first day of VL-1 operations (the day of its landing). {Ed.}

[6] Taken from Viking: The Exploration of Mars. NASA EP-208 [JPL 400-219 5/84; U.S. GPO: 1984-784-698]. {Ed.}

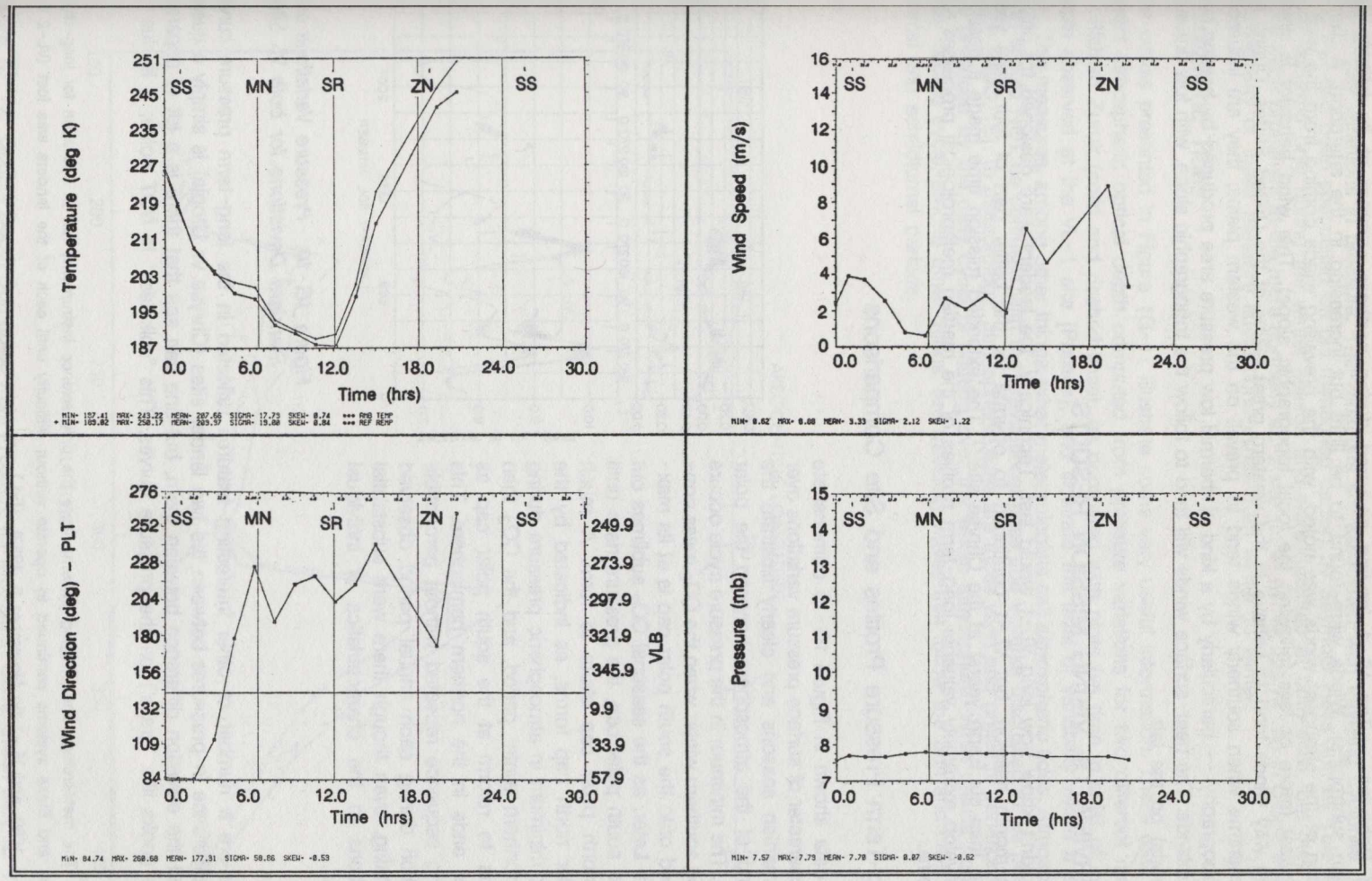


Figure 1G-9: Sol-1 VL-1 Meteorology Data
(top, left/right) a. Temperature, b. Wind Speed, (bottom, left/right) c. Wind Direction, d. Atmospheric Pressure

Wind Behavior -- The VL-1 landing occurred during northern summer at its latitude (the martian subtropics). Winds were found to be light but increasing in the afternoon. A diurnal rotation of the strongest winds was noted, with the prevailing winds coming from the west to southwest (more or less following the local topographic slope). This wind behavior is similar to the way winds tend to behave in the western great plains of the United States. During the summertime when southerly winds tend to prevail on the western plains, they are influenced by topography -- particularly by a kind of thermal low pressure area produced by heating over the uplands. The near-surface winds will tend to follow the topographic slope, with low pressure (uplands) on the left.

OVERVIEW OF VIKING MISSION RESULTS

We didn't know how long VL-1 would last. Technically, the landers were designed for only a three-month mission, but VL-1 continued to operate for the better part of four Mars years (more than six Earth years) at the Chryse site[7]. The extended mission time made it possible to develop extremely valuable long-term profiles of the martian meteorological processes and systems.

Long-Term Pressure Profiles and Site Comparisons

The data shown in Figure 1G-10 demonstrate the character of surface pressure variations over the martian seasons and clearly illustrate the cycling of the atmosphere between the polar caps. The minimum in the pressure cycle occurs during southern winter when the CO_2 mass condensed onto the south polar cap is at its maximum. Later, as the seasonal CO_2 sublimes out of the south polar cap, the pressure rises until the north polar cap starts to form. A much smaller north cap forms, as indicated by the weak minimum in atmospheric pressure during the northern winter period, and the CO_2 then begins to reform at the south polar cap as winter ends in the northern hemisphere. This cycling sequence repeated without perceptible variation during each annual period observed by Viking, even though there were substantial variations in the characteristics of individual cycles.

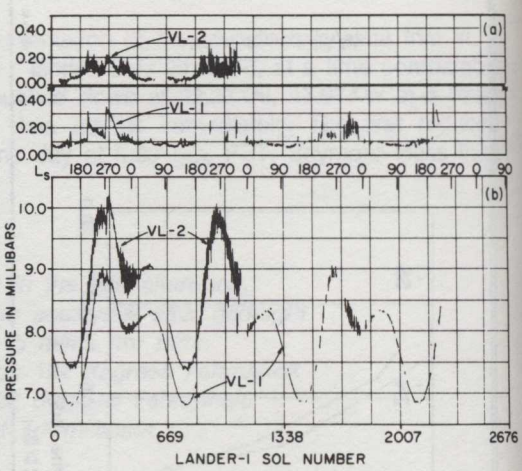


Figure 1G-10: Pressure Variations and Standard Deviations for both VL Sites

There are a number of other interesting features reflected in the long-term pressure curve. The difference in pressures between the two lander sites (Chryse vs Utopia) is simply a question of the elevation difference between them, but one can see that there is a lot of apparent noise (looks like "grass") on the pressure curves. This "spikiness" is *NOT* noise; it is due to

---

[7] The meteorology and imaging systems were the only science instruments left in operation for long-term science, and these systems continued to operate without difficulty until each of the landers was lost (VL-2 on April 11, 1980, and VL-1 on November 5, 1982). {Ed.}

day-to-day variations in pressure. The variations are associated with traveling cyclones of the kind that had been speculated on (as mentioned earlier) by Seymour Hess and by Yale Mintz. They occurred only during the winter season and were detected at both sites, but they were much more prominent at the more northerly VL-2 site (at 47.67° N, 225.74° W) in Utopia Planitia. The intensity of these pressure fluctuations varies from winter to winter.

Atmospheric Tides

The curves presented in Figure 1G-11 illustrate some very useful information. These data represent atmospheric optical depth computed from pressure variations for two different models by Richard Zuruk (solid and dashed lines), as compared with plots (vs time in sols) of optical depth observed at the VL-1 site [Pollack], and illustrate the level of variability within a single day. This measures, among other things, what meteorologists call *atmospheric tides*. Atmospheric tides should NOT be confused with the gravitational tides Joe Veverka discussed in his presentation [NMC-1E]. In meteorology, atmospheric tides are those wind and pressure variations that are produced by the daily cycle of heating over the whole atmosphere. What results from the daily loading cycle, among other things, are traveling waves that follow the Sun and have both diurnal and semidiurnal periods.

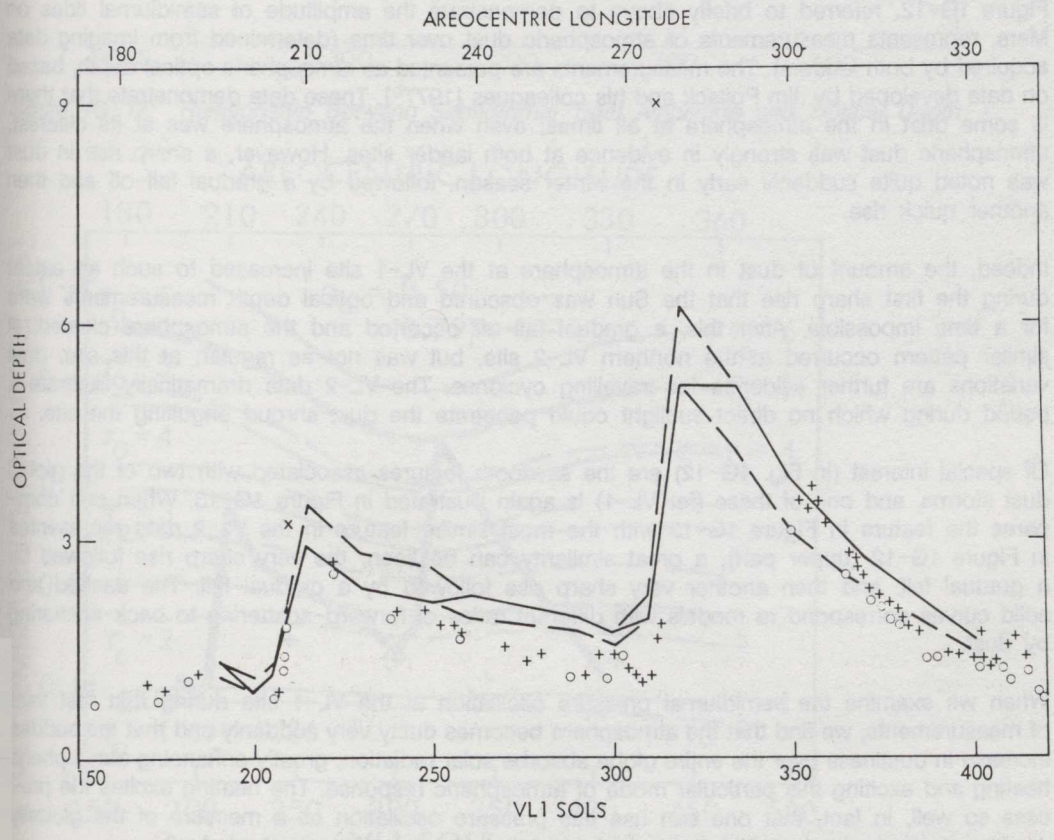


Figure 1G-11: Optical Depth Computed from Pressure for Models in Comparison with VL-1 Data

One can see marked variations in the amplitudes of these tides. The variations are reflected in patterns, such as the repeated sharp rise and gradual fall reflected in both the modeled curve and during the first VL-1 year. Such variations are associated mainly with the semidiurn component of the atmospheric tide. As reflected in Figure 1G-12, one can see in the Viking da (recorded over time by both landers) that there are substantial variations in the atmospher tides from winter to winter, and that the largest tides (the largest daily variations) occur durir the northern winter season -- the period when Mars is closest to the Sun (perihelion). Th solid and dashed curves correspond to models with different scattering ratios (forward-to-bac by dust.

MARTIAN DUST STORMS

I want to move now to a discussion of martian dust storms, and in particular to what wa learned from various Viking instruments about storm characteristics. I am going to focus on global dust storms. Mars in fact experiences many localized dust storms, and many were observed by the two Viking orbiters. But the most spectacular meteorological events are those large storms during which the atmosphere over essentially the entire planet appears to fill with dust for periods of at least a few weeks or longer.

Atmospheric Dust and Optical Depth

Figure 1G-12, referred to briefly above to demonstrate the amplitude of semidiurnal tides or Mars, represents measurements of atmospheric dust over time (determined from imaging data acquired by both landers). The measurements are presented as atmospheric optical depth, based on data developed by Jim Pollack and his colleagues [1977[b]]. These data demonstrate that there is some dust in the atmosphere at all times; even when the atmosphere was at its clearest atmospheric dust was strongly in evidence at both lander sites. However, a sharp rise in dust was noted quite suddenly early in the winter season, followed by a gradual fall-off and then another quick rise.

Indeed, the amount of dust in the atmosphere at the VL-1 site increased to such an extent during the first sharp rise that the Sun was obscured and optical depth measurements were for a time impossible. After this, a gradual fall-off occurred and the atmosphere cleared. A similar pattern occurred at the northern VL-2 site, but was not as regular; at this site, d variations are further evidence for travelling cyclones. The VL-2 data dramatically illustrate period during which no direct sunlight could penetrate the dust shroud engulfing the site.

Of special interest (in Fig. 1G-12) are the sawtooth features associated with two of the global dust storms, and one of these (for VL-1) is again illustrated in Figure 1G-13. When one compares the feature in Figure 1G-13 with the most similar feature in the VL-2 data represented in Figure 1G-12 (upper part), a great similarity can be seen: the very sharp rise followed a gradual fall, and then another very sharp rise followed by a gradual fall. The dashed and solid curves correspond to models with different ratios of forward-scattering to back-scatterin by dust.

When we examine the semidiurnal pressure oscillation at the VL-1 site during that first yea of measurements, we find that the atmosphere becomes dusty very suddenly and that the sudden increase in dustiness over the entire globe absorbs solar radiation, greatly enhancing atmospheri heating and exciting this particular mode of atmospheric response. The heating excites the pro cess so well, in fact, that one can use this pressure oscillation as a measure of the globally varying atmospheric dust (the time-varying, globally-averaged amount of dust).

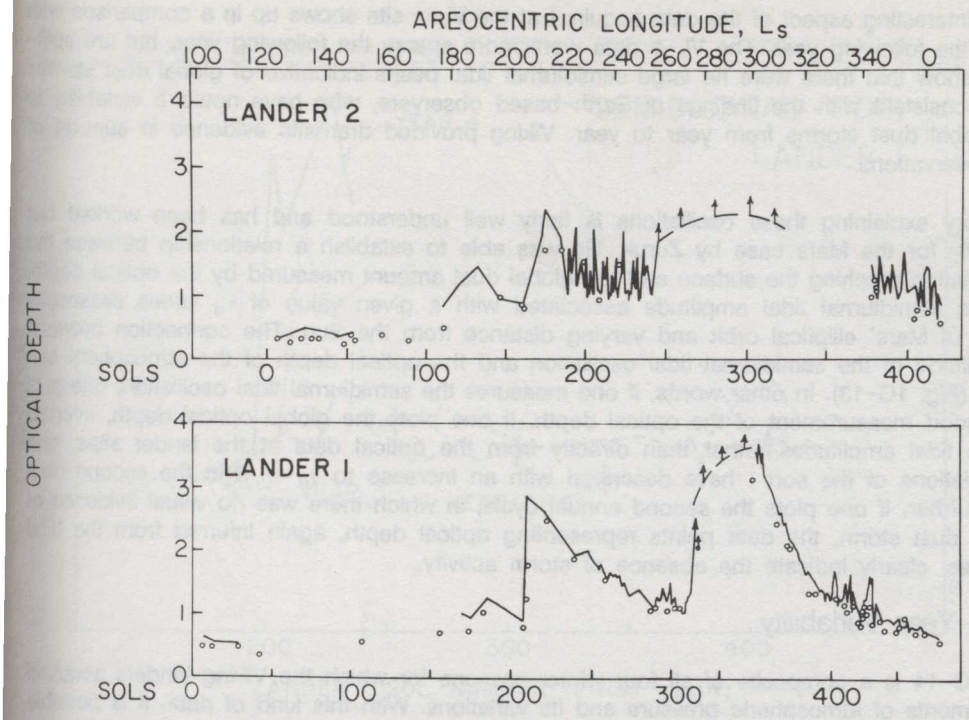


Figure 1G-12: Relationship Between Semidiurnal Tidal Amplitude and Optical Depth

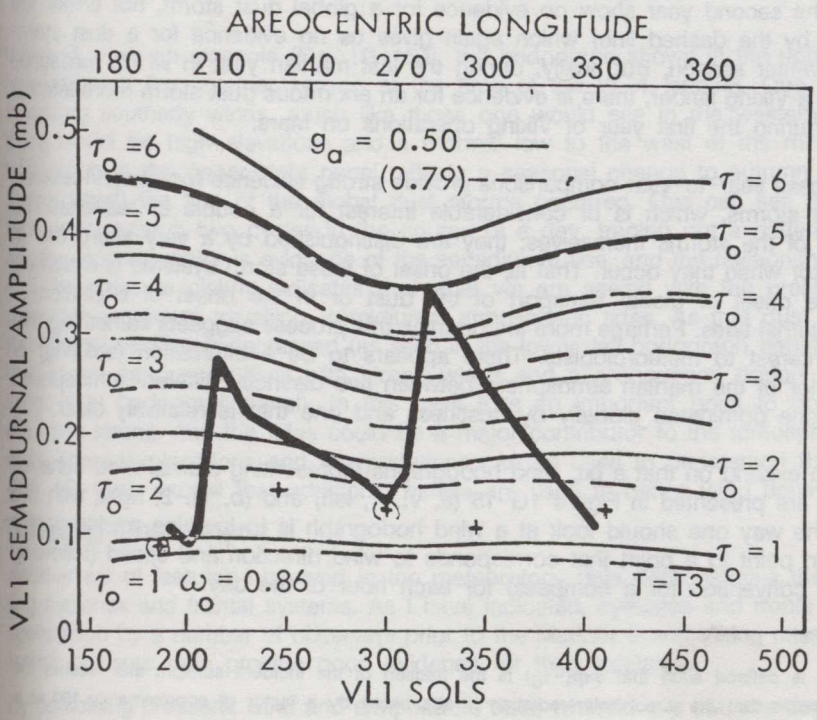


Figure 1G-13: Detail of "Sawtooth" Variation in Tides

Another interesting aspect of the data acquired at the VL-1 site shows up in a comparison with data for the following year. The VL-1 data were more sparse the following year, but are sufficient to show that there were no large semidiurnal tidal peaks indicative of global dust storms. This is consistent with the findings of Earth-based observers, who have noted a variability in these global dust storms from year to year. **Viking** provided dramatic evidence in support of those observations.

The theory explaining these oscillations is fairly well understood and has been worked out specifically for the Mars case by Zuruk. He was able to establish a relationship between the solar radiation reaching the surface and the global dust amount measured by the optical depth, τ_d[8]. The semidiurnal tidal amplitude associated with a given value of τ_d varies seasonally because of Mars' elliptical orbit and varying distance from the Sun. The connection between the amplitude of the semidiurnal tidal oscillation and the optical depth of the atmosphere can be seen (Fig. 1G-13). In other words, if one measures the semidiurnal tidal oscillation, one gets a very good measurement of the optical depth. If one plots the global optical depth, inferred from the tidal amplitudes rather than directly from the optical data at the lander sites, one sees variations of the sort I have described with an increase to $\tau_d = 5$ in the second dust storm. Further, if one plots the second annual cycle, in which there was no visual evidence of a global dust storm, the data points representing optical depth, again inferred from the tidal amplitudes, clearly indicate the absence of storm activity.

Year-to-Year Variability

Figure 1G-14 is a composite of all four winter seasons for which the **Viking** landers acquired measurements of atmospheric pressure and its variations. With this kind of data, it is possible to say a little bit more about the interannual variability. The first year--the one with what we can think of as two calibrating storm systems--is represented by the dotted curve. The data points representing the second year show no evidence for a global dust storm, not unlike the third year (indicated by the dashed line) which again gives us no evidence for a dust storm during the northern winter season. But finally, during the last martian year in which measurements were made by a **Viking** lander, there is evidence for an enormous dust storm more intense than anything seen during the first year of **Viking** operations on Mars.

Characteristics -- These year-to-year comparisons provide strong evidence for interannual variability in global dust storms, which is of considerable interest for a couple of reasons. The first is the character of the storms themselves; they are distinguished by a very sharp rise in the optical depth factor when they occur. That is, the onset of these storm systems is extremely sudden, either in the onset of global transport of the dust or in the onset of excitation of the associated semidiurnal tides. Perhaps more importantly, this process suggests something that is of even greater interest to meteorologists. There appears to be a bifurcation (splitting) of the dynamical behavior of the martian atmosphere between two distinctly different atmospheric circulation regimes, one dominated globally by dustiness and one that is relatively clear.

Wind Behavior -- To expand on that a bit, wind hodographs representing diurnal wind behavior at both landing sites are presented in Figure 1G-15 (*a*, VL-1, left) and (*b*, VL-2, right, with two successive years). The way one should look at a wind hodograph is to imagine a wind vector drawn from the origin point to a point that corresponds to wind direction and speed (following the usual directional convention for a compass) for each hour of the day.

---

[8] Optical depth τ_d is defined such that $\exp(-\tau_d)$ is the fraction of the incident sunlight that reaches the surface. Thus $\tau_d = 5$ means that the direct solar radiation is attenuated by a factor of approximately 150 as a result of the dust in the atmosphere.

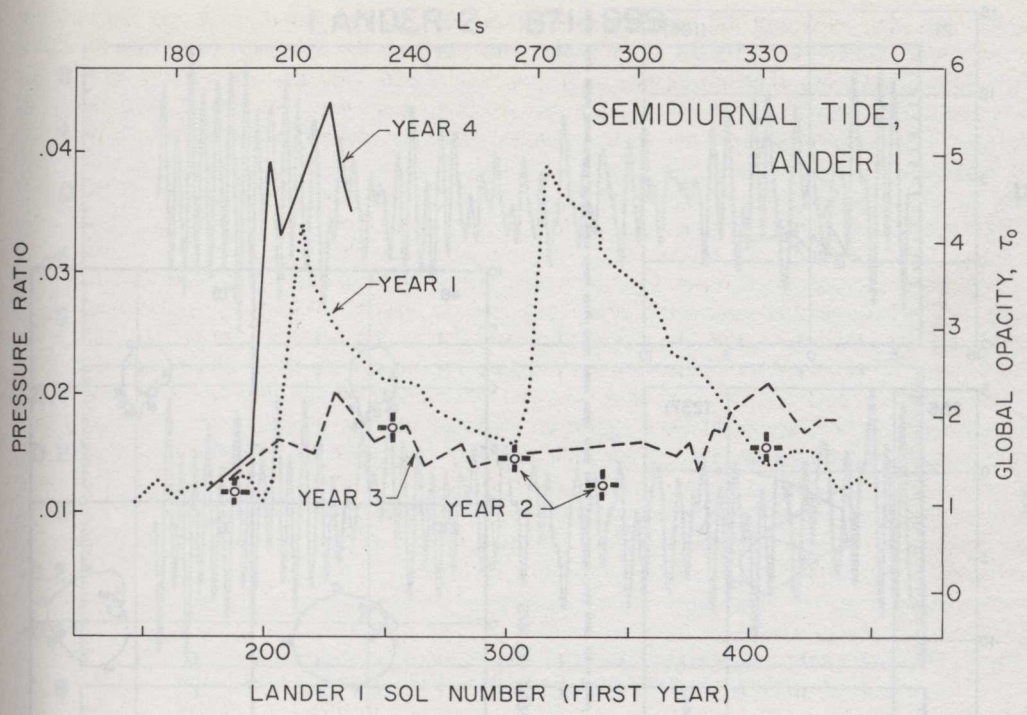


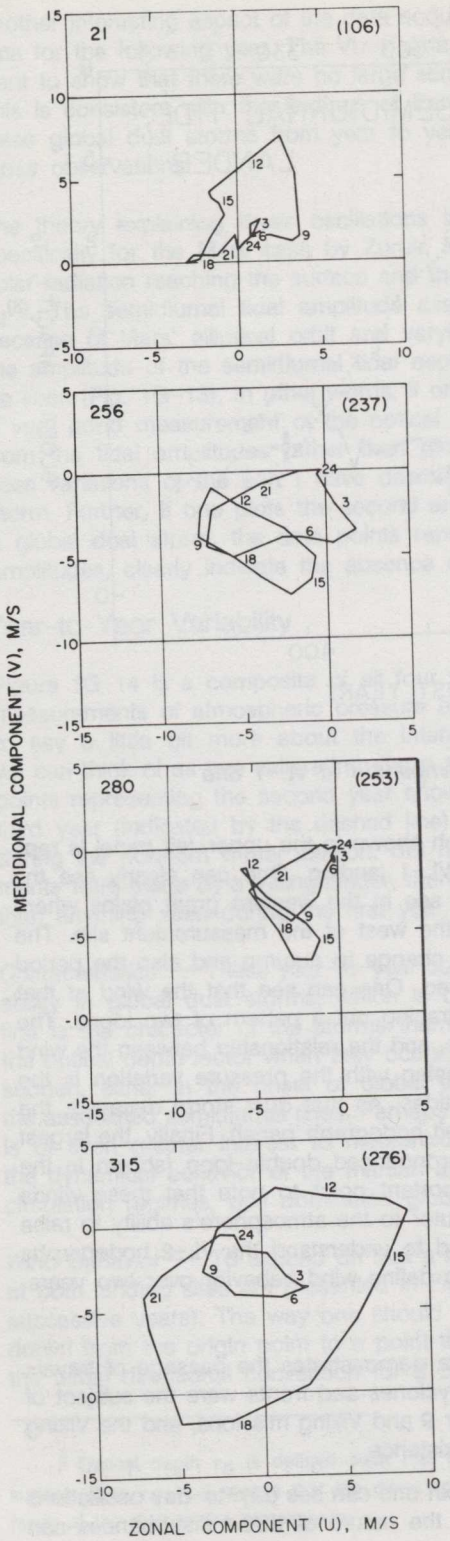
Figure 1G-14: Four-Year Composite of Semidiurnal Amplitude at VL-1 Site

Using VL-1 as an example (Fig. 1G-15a), the hodograph shown in the upper-left panel is representative of the summer season at the time of the VL-1 landing. One can clearly see the prevailing southerly winds, much like those one would see in the western great plains where there would be high elevations and a thermal low to the west of the measurement site. The hodograph in the upper-right panel reflects a seasonal change to autumn and also the period during which the first of the global dust storms occurred. One can see that the wind at that time went through two cycles in the course of a day, tracing out a pattern of two loops. The double-loop signature is evidence of the semidiurnal tide, and the relationship between the wind and the pressure clearly indicates that what we are seeing with the pressure variation is the planetary, westward travelling, semidiurnal atmospheric tides. As that dust storm decayed, the semidiurnal oscillation decreased (as seen in the lower-left hodograph panel). Finally, the largest dust storm announces itself with strong winds and a pronounced double-loop (shown in the lower-right hodograph panel). In this case it is an important point to note that these winds are quite strong, and the tides could be a major contributor to the atmosphere's ability to raise dust. These explanations and interpretations can be used to understand the VL-2 hodographs (Fig. 1G-15b); except that additional panels are used to define wind behavior over two years.

Travelling Phenomena

Another set of features observed in the meteorology data demonstrates the passage of travelling cyclones and frontal systems. As I have indicated, cyclones and fronts were the subject of speculation by a number of observers prior to the **Mariner 9** and **Viking** missions, and the **Viking** lander pressure data provide good evidence for their existence.

By combining pressure, wind and temperature data, wherein one can see day-to-day oscillations in their variables (as shown for VL-2 in Figure 1G-16), the nature of these disturbances can



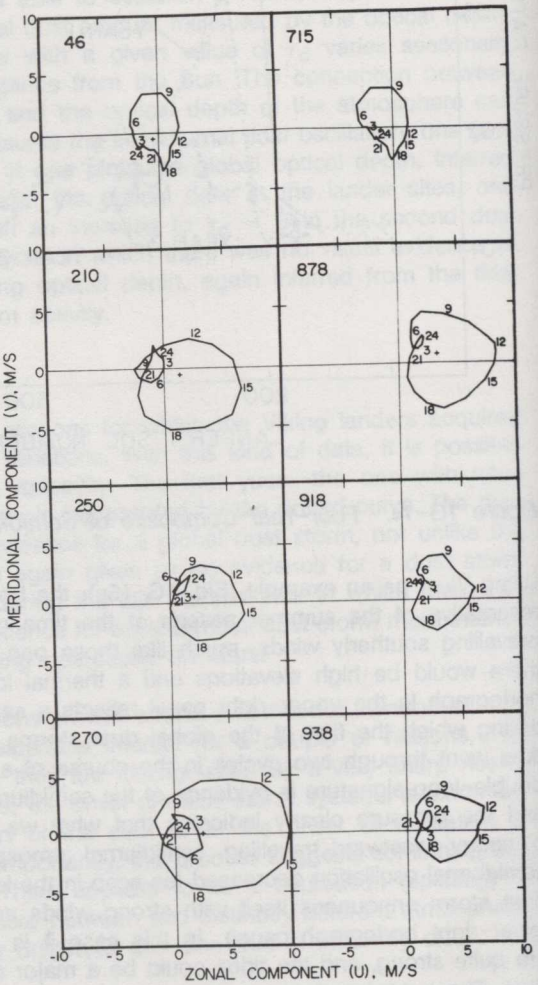


Figure 1G-15:

Wind Hodographs Reflecting Wind Velocity/Direction Patterns at Viking Lander Sites

- *Left Column - VL-1*
- *Above - VL-2*

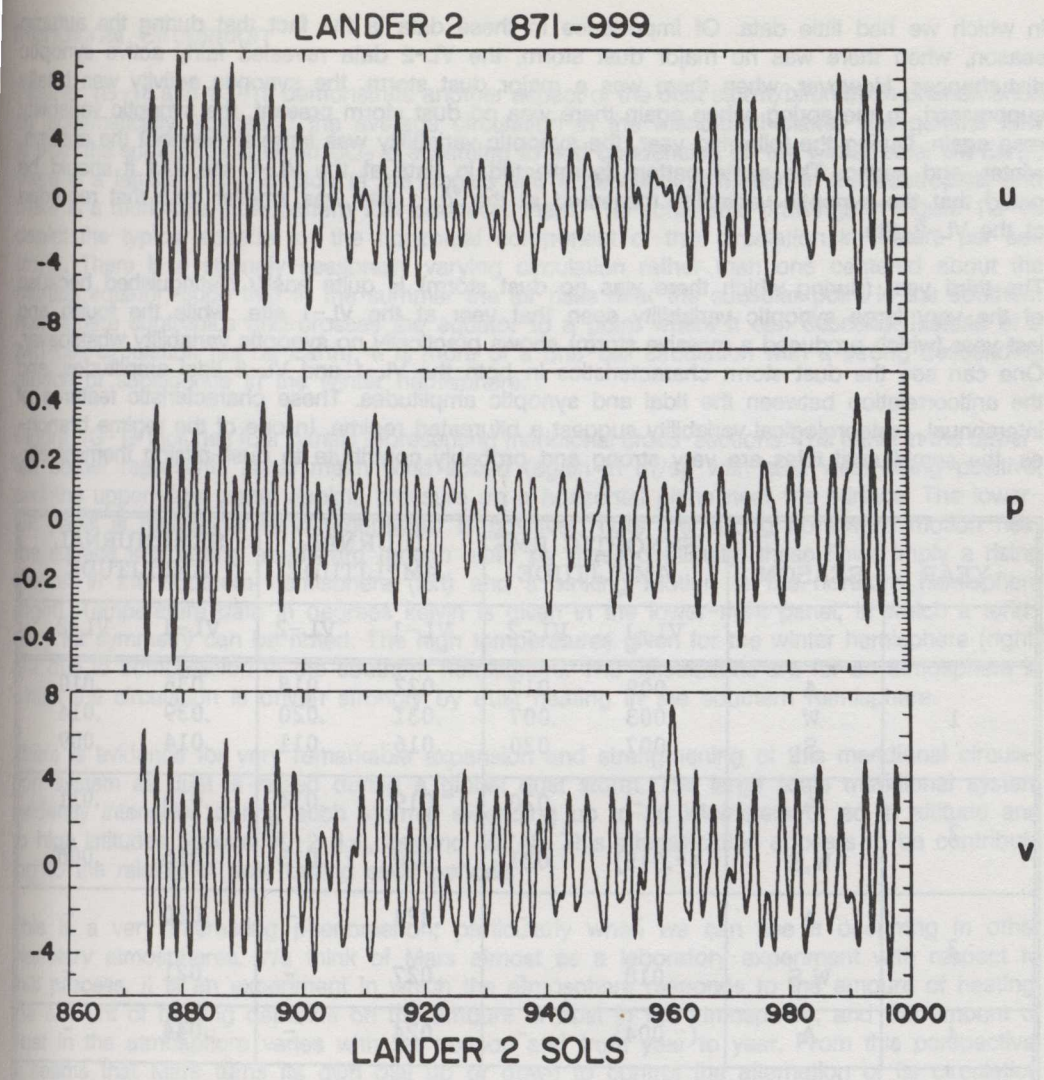


Figure 1G-16: Pressure and Wind Data for VL-2; u = *West to East Wind,* p = *Pressure,* v = *Temperature Data*

be deduced. Note that the maximum negative meridional wind (the maximum wind from the north) precedes just slightly the highest pressure point in each oscillation. That is typically what we see on Earth when systems travel from west to east. It indicates travelling wave cyclones moving from west to east in the strong westerly winds of Mars' northern hemisphere. This is very similar to the behavior of such features in Earth's atmosphere. We can even measure and assess the spatial scales of these martian systems: there are about two to four waves moving around the planet in the westerly stream, and the mid-latitude VL-2 site lies in its path.

Figure 1G-17 is a table of some data that are indicative of the year-to-year variations in these travelling waves, as well as the variation in the tides that I have already discussed. All four years measured by Viking are represented according to season, beginning with: autumn (A) and progressing to winter (W) and spring (S). Winter and spring are combined during that period

in which we had little data. Of importance in these data is the fact that during the autumn season, when there was no major dust storm, the VL-2 data revealed fairly active synoptic disturbances. However, when there was a major dust storm, the synoptic activity was totally suppressed. In the spring, when again there was no dust storm present, the synoptic variability rose again. During the following year, the synoptic variability was large throughout the autumn, winter, and spring. The same pattern is reflected in data at the VL-1 site, but it should be noted that the synoptic variability measured at the VL-1 site was smaller than that recorded at the VL-2 site.

The third year (during which there was no dust storm) is quite easily distinguished because of the very large synoptic variability seen that year at the VL-1 site, while the fourth and last year (which produced a massive storm) shows practically no synoptic variability whatsoever. One can see the dust storm characteristics in both the VL-1 and VL-2 tidal amplitudes, and the anticorrelation between the tidal and synoptic amplitudes. These characteristic features of interannual, meteorological variability suggest a bifurcated regime. In one of the regime branches, the semidiurnal tides are very strong and probably contribute to dust-raising themselves.

YEAR	SEASON	SYNOPTIC RMS AMPLITUDE		DIURNAL AMPLITUDE		SEMIDIURNAL AMPLITUDE	
		VL-1	VL-2	VL-1	VL-2	VL-1	VL-2
1	A	.008	.017	.022	.018	.036	.010
	W	.003	.007	.031	.020	.039	.014
	S	.007	.020	.016	.011	.014	.009
2	A	-	.020	.019	.014	.015	.011
	W,S	(~.015)	.021	.020	.013	.016	.006
3	A	-	-	.024	-	.020	-
	W,S	.018	-	.027	-	.021	-
4	A	(~.004)	-	.024	-	.044	-

A = Autumn, W = Winter, S = Spring

Figure 1G-17: Table of Pressure Variations

As they become very strong, the atmospheric circulation (as a whole) intensifies except for the synoptic features; these travelling cyclone waves were suppressed, at least at the latitudes where we measured them.

This leaves the question of the initiation mechanism for the global dust storms unanswered. Why, for example, do they occur in some years and not in others? One is also led to many new questions connected with the occurrence of these systems which must reflect major factors in modifying the martian surface through wind erosion and particle transport. One is led to wonder, for example, about how global dust storms, atmospheric tides, and synoptic waves may have varied in the past with different orbital parameters or different mean surface pressures.

Meridional Circulation

Figures 1G-18 and 1G-19 demonstrate another aspect of the dust storm/bifurcation phenomenon by schematically illustrating the average circulation in the meridional plane. The general flow is from the summer pole when CO_2 is subliming to and condensing on the winter pole. On Earth, there is a familiar rising motion in the tropics and a descending motion in the subtropics, and there is a meridional flow pattern that connects them. The schematic drawings in Figure 1G-18 depict the typical speeds for the horizontal component of this circulation in meters per sec (m/s). There is a strongly-seasonally varying circulation rather than one centered about the martian equator, such that in the summer the air rises near the subsolar point in the southern hemisphere subtropics and crosses the equator to a point where it can descend. Instead of a two-cell circulation (as on Earth), it is more of a one-cell circulation with a strong descending motion or subsidence in the winter hemisphere.

Figure 1G-19 includes four panels representing meridional cross-sections. The panel in the upper-left corner represents zonal mean wind speed (again in m/s), with east-west being positive, and the upper-right panel depicts pressure on a horizontal plane near the surface. The lower-left panel is meridional wind (southward is positive); note the strong northward motion near the surface and strong southward motion aloft. By mass continuity, these flows imply a rising motion in the southern hemisphere (left) and a sinking motion in the northern hemisphere (right). Temperature data in degrees kelvin is given in the lower-right panel, in which a tendency for symmetry can be noted. The high temperatures given for the winter hemisphere (right) are due to compression in the southern hemisphere. The calculations are for an atmosphere in which the circulation is driven strongly by dust heating in the southern hemisphere.

There is evidence for very remarkable expansion and strengthening of this meridional circulation system as dust is raised during a global dust storm. The large scale meridional system evidently intensifies during such storms, extending up to 50 kilometers or so in altitude and to high latitudes beyond VL-2 (i.e., beyond 50°N). This intensification appears to be contributing to the raising of dust during such periods.

This is a very interesting phenomenon, particularly when we can see it occurring in other planetary atmospheres. We think of Mars almost as a laboratory experiment with respect to this process. It is an experiment in which the atmosphere responds to the amount of heating, the amount of heating depends on the amount of dust in the atmosphere, and the amount of dust in the atmosphere varies with the season and from year to year. From this perspective, it seems that Mars turns its own dial up or down to control the alternation of its circulation regimes.

A significant impact of this enhanced meridional circulation is that the same kind of feature has been observed in terrestrial general-circulation model studies, and has been demonstrated in simulations of the **nuclear winter** phenomenon. Given enough atmospheric absorbing material (smoke and dust) to strongly and sufficiently absorb solar radiation, a greatly enhanced meridional circulation analogous to that observed on Mars can develop quite suddenly on Earth. The nuclear winter simulations suggest a bifurcation between weakly absorbing and strongly absorbing circulation regimes.

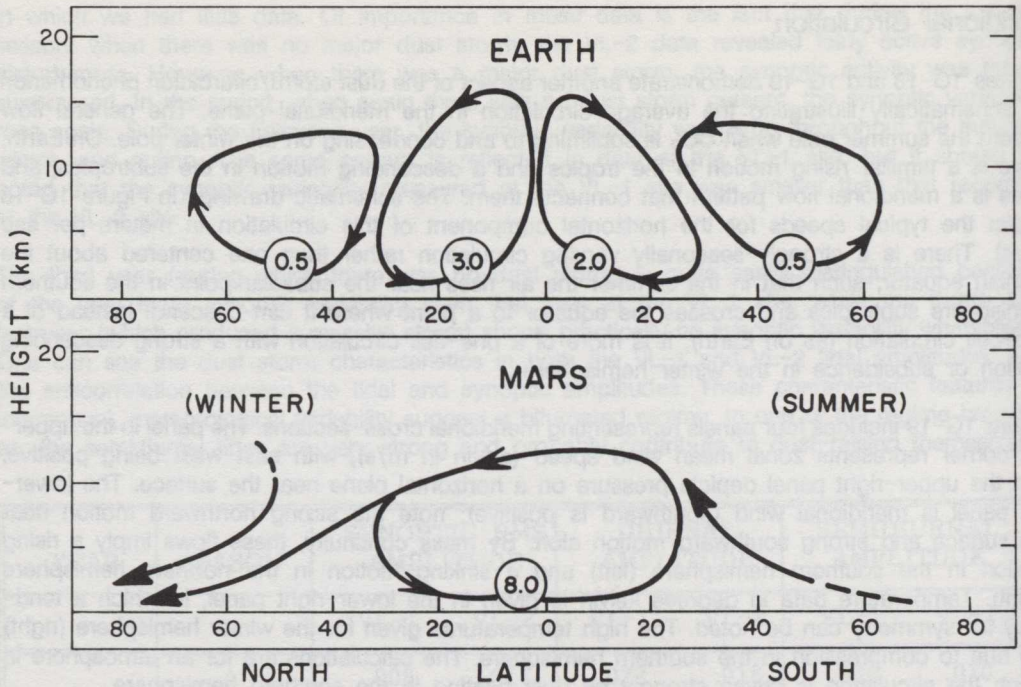


Figure 1G-18: Schematic of Meridional Winds in Meridional Plane

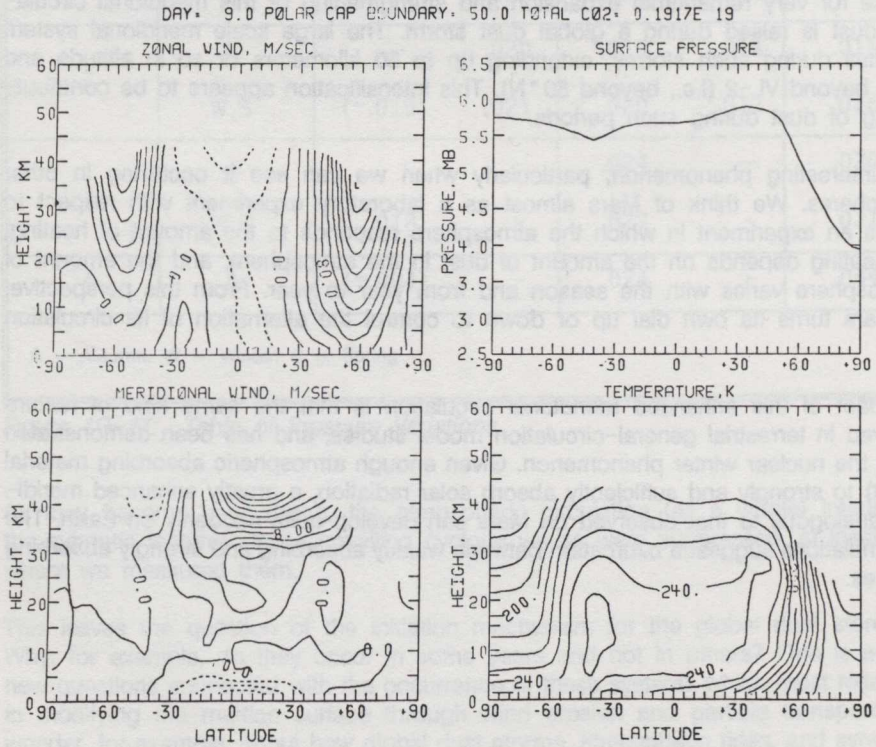


Figure 1G-19: Meridional Plane Cross Section Calculations [Haberle]

CONCLUSION

There is much that we would yet like to learn about the meteorology of Mars. In particular, of course, there is considerable interest in the cycling of water vapor between that atmosphere and the surface. The **Mars Observer** mission represents, perhaps for the first time, an orbiter mission that is ideally suited to making synoptic atmospheric measurements. Its data products will allow us to look at Mars in a manner that will facilitate good transport measurements, and will allow us to begin to assess the sources and sinks for water vapor. ■

REFERENCES

References identified here by alphabetically sequenced lower-case characters are keyed to *specific* references in the presentation text where the same characters are used as identification tags. In the text, however, the identifier will be found within brackets, either by itself or with a name and/or credit date (i.e., when formally published) -- e.g., [a] or [1978a].

This space is otherwise available for notes.

~~~~~~~~

UNSPECIFIED REFERENCES

Hess, S.L. 1973. Martian winds and dust clouds. *Planet. Space Sci.* 21:1549-1557.

Hess, S.L., Henry, R.M., Tillman, J.E. 1979. The seasonal variation of atmospheric pressure on Mars as affected by the south polar cap. *J. Geophys. Res.* 84:2923-27.

Hess, S.L., Henry, R.M., Leovy, C.B., Ryan, J.A., and Tillman, J.E. 1977. Meteorological results from the surface of Mars: Viking 1 and 2. *J. Geophys. Res.* 82:4559-74.

Hess, S.L., Ryan, J.A., Tillman, J.E., Henry, R.M., and Leovy, C.B. 1980. The annual cycle of pressure on Mars measured by Viking 1 and 2. *Geophys. Res. Lett.* 7:197-200.

Leovy, C.B. 1977. The atmosphere of Mars. *Sci. Am.* 237:34-43.

Leovy, C.B., and Mintz, Y. 1969. Numerical simulation of the weather and climate of Mars. *J. Atmos. Sci.* 26:1167-90.

Leovy, C.B., and Zuruk, R.W. 1979. Thermal tides and martian dust storms: Direct evidence for coupling. *J. Geophys. Res.* 84:2956-68.

PRESENTATION REFERENCES

a. Pollack, J.B., Leovy, C.B., Mintz, Y.H. and Van Camp, W. 1976. Winds on Mars during the Viking season: Predictions based on a general circulation model with topography. *Geophys. Res. Lett.* 3:479-82.

b. Pollack, J.B., Colburn, D., Kahn, R., Hunter, J., Van Camp, W., Carlston, C.E., and Wolf, M.R. 1977. Properties of aerosols in the martian atmosphere, as inferred from Viking lander imaging data. *J. Geophys. Res.* 82:4479-96.
~~~~~~~~

THE NASA MARS CONFERENCE

Edited by Duke B. Reiber

SCIENCE AND TECHNOLOGY SERIES

523.43
N241

LOS ANGELES PUBLIC LIBRARY
3 7244 1640 1674 4